Exercises

Problem 6.1: Ramsey measurement

Consider a single qubit with energy splitting ω and Hamiltonian $\hat{H}$

$$\hat{H} = \frac{1}{2}\omega\hat{\sigma}_z. \tag{6.1}$$

We consider a Ramsey measurement for estimating ω. The Ramsey sequence consists of the steps: *(i)* initialise the qubit in state $|0\rangle$, *(ii)* apply the rotation $\hat{R}_y(\frac{\pi}{2})$, i.e. a $\frac{\pi}{2}$ pulse (c.f. Eq. (2.53)), *(iii)* let the qubit evolve freely under $\hat{H}$ for a time t, *(iv)* apply the inverse rotation $\hat{R}_y(-\frac{\pi}{2})$, and *(v)* measure in the $\{|0\rangle, |1\rangle\}$ basis.

(a) Illustrate each step *(i)-(v)* on the Bloch sphere.

(b) Show that the probability of the outcome corresponding to $|1\rangle$ is given by

$$p_1 = \sin^2(\frac{\omega t}{2}). \tag{6.2}$$

(c) Show that the Fisher information of the distribution of outcomes, with respect to ω, is

$$\mathcal{F}_\omega = t^2. \tag{6.3}$$

How long should you measure to get the best precision?

(d) Consider now that noise will cause the qubit to decay to its ground state with relaxation time T_1, such that p_1 is modified to

$$p_1 = \sin^2(\frac{\omega t}{2})e^{-t/T_1}. \tag{6.4}$$

Compute the Fisher information in this case. What is the optimal measurement time now?

Problem 6.2: Rabi measurement with non-zero free Hamiltonian

Consider a Rabi measurement on a single qubit consisting of *(i)* initialisation in $|0\rangle$, *(ii)* evolution for time t under the Hamiltonian $\hat{H} = \frac{1}{2}\omega_x\hat{\sigma}_x + \frac{1}{2}\omega_z\hat{\sigma}_z$ and, *(iii)* a projective measurement in the z-basis, $\{|0\rangle, |1\rangle\}$.

(a) Show that the probability for the outcome corresponding to $|1\rangle$ is given by

$$p_1 = \frac{\omega_x^2}{\omega_x^2 + \omega_z^2}\sin^2(\frac{1}{2}\sqrt{\omega_x^2 + \omega_z^2}\, t). \tag{6.5}$$

(b) Plot the corresponding Fisher information $\mathcal{F}_{\omega_x}$ with respect to the transverse frequency ω_x as a function of time, experimenting with different regimes for the frequencies ($\omega_x < \omega_z$, $\omega_x \approx \omega_z$, $\omega_x > \omega_z$).

(c) Argue that the scaling of the optimal $\mathcal{F}_{\omega_x}$ with time is quadratic, t^2.